

# Stack Overflow Developer Survey

Sabrina Götz, 10.08.2025



© IBM Corporation. All rights reserved.

# OUTLINE

---



- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

# EXECUTIVE SUMMARY

---



- Modern, open, and cloud-based technologies are on the rise
- Python is set to surpass JavaScript in popularity
  - Rust and Go are emerging as strong players
  - JavaScript remains core for web development despite slight decline
  - C is losing relevance in this audience
- PostgreSQL strengthens its lead among databases
  - SQLite is becoming increasingly important, presumably because of simple use cases
  - Supabase enters the top 10, reflecting cloud-native adoption
  - Oracle drops out, showing reduced interest in complex proprietary systems
- Cloud platforms and modern frameworks remain in high demand
- A young, well-educated respondent base is driving rapid adoption of new technologies



# INTRODUCTION

---



- Purpose:
  - Analyzing current and future technology trends based on survey data
  - Focus: Programming Languages, Databases, Web Frameworks and Cloud Platforms
- Target Audience:
  - Developers, Data Analysts, and other Professionals who actively use these technologies
  - Industry stakeholders shaping organizational technology adoption
- Value:
  - Developers, Data Analysts, and other Professionals applying these technologies in their work
  - Industry stakeholders shaping and investing in technology adoption.



# METHODOLOGY

---

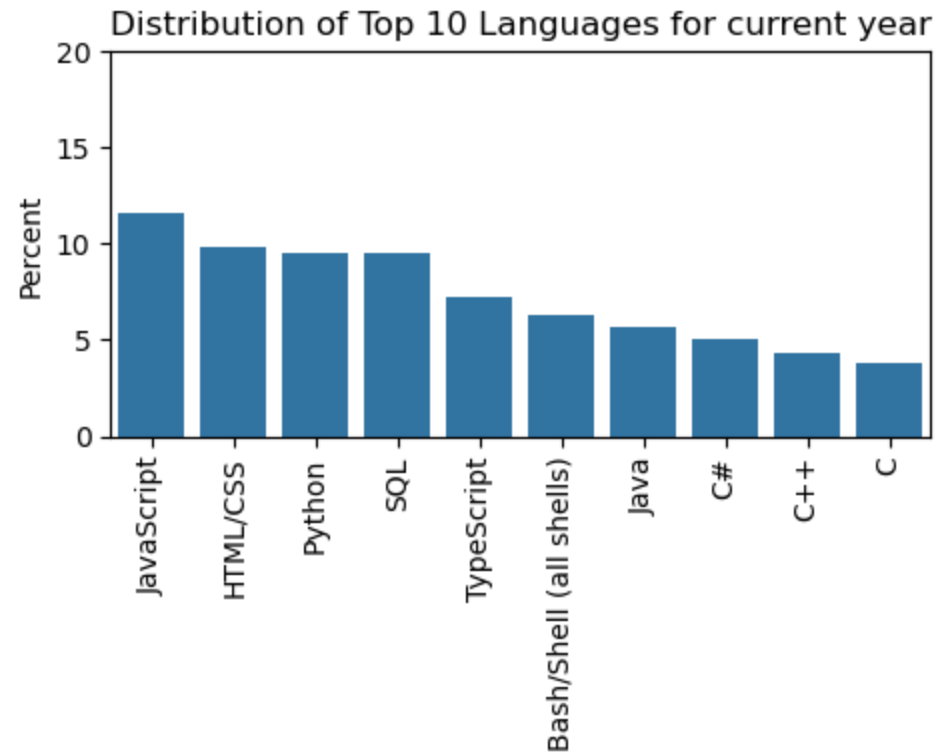


- **Data Source:**
  - The dataset used in this study originates from the Stack Overflow Developer Survey, which is conducted annually by Stack Overflow.
- **Collection Method:**
  - The data is provided in CSV format, which can be downloaded together with a detailed survey schema
- **Key Wrangling Steps**
  - Removed 435 duplicate records
  - Filled missing values in single-choice fields with the mode; forward-filled sequential fields
  - Split multi-select responses into

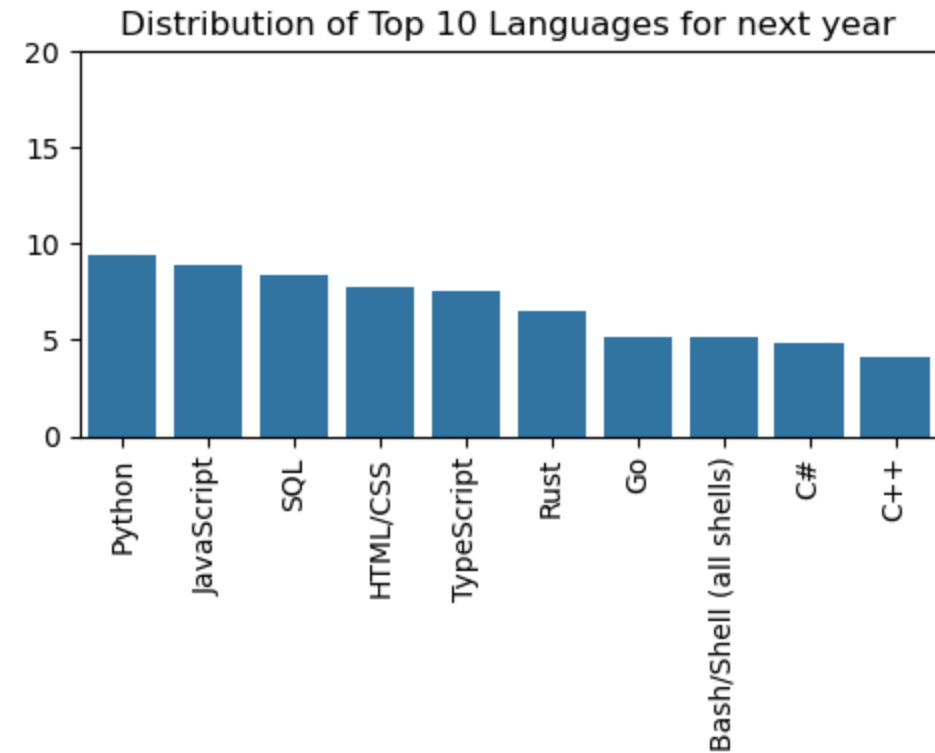


# PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



# PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

---

## Findings

- JavaScript currently leads (~11%), but is losing ground slightly.
- Python is currently behind JavaScript and HTML/CSS (~9.5%), but is expected to take the lead next year (~9.3%).
- Rust and Go are currently missing from the top 10, but are gaining relevance.
- C currently stands at ~3.7%, but will fall out of the top 10 next year.

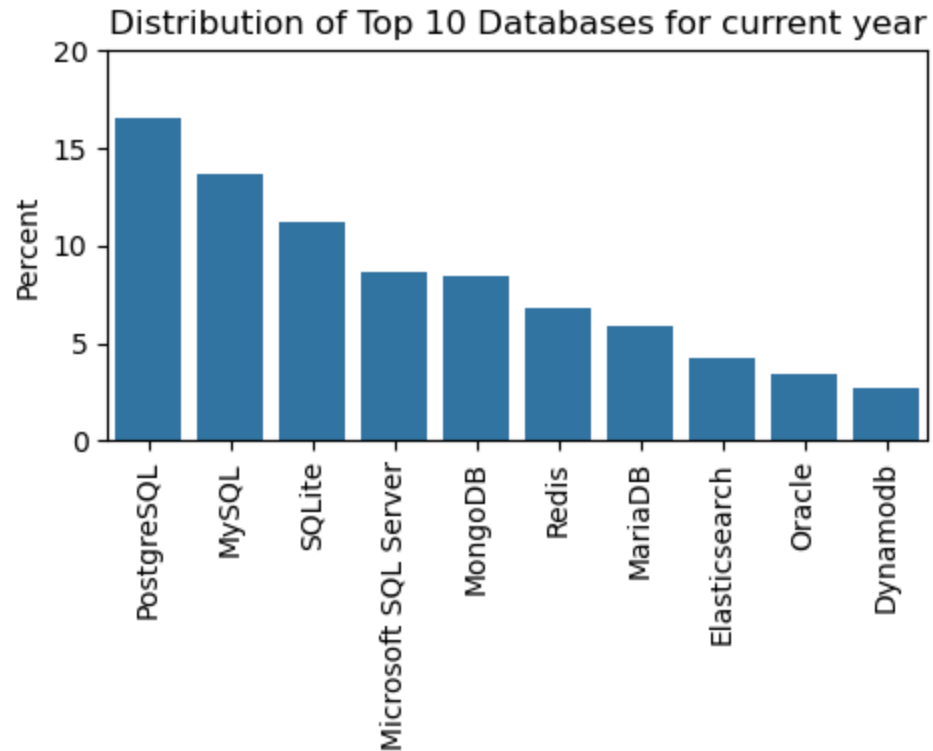
## Implications

- JavaScript remains a core technology for the web, despite a slight decline.
- Python benefits from data science, AI, and automation—skills are worthwhile.
- Rust and Go are expected to gain market share, especially in system, cloud, and high-performance apps.
- C is losing importance in this target group but remains relevant in niche areas.

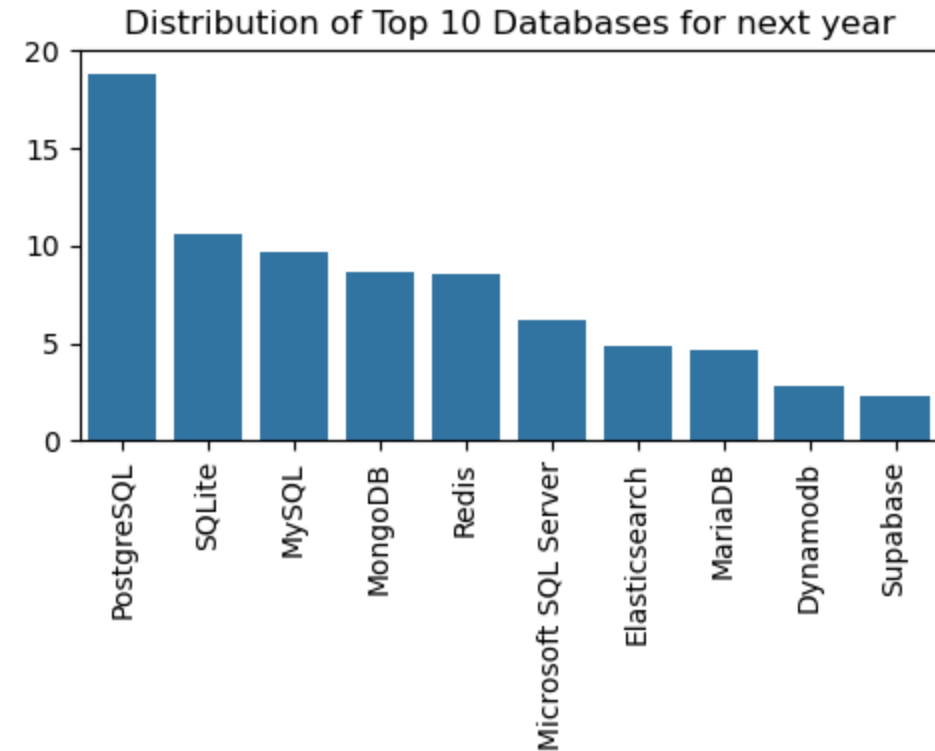


# DATABASE TRENDS

Current Year



Next Year





# DATABASE TRENDS - FINDINGS & IMPLICATIONS

---

## Findings

- PostgreSQL is currently the leader (~16%) and will continue to gain market share (~19%) by next year.
- SQLite is rising significantly in the forecast and overtaking MySQL.
- Oracle is disappearing from the top 10 for next year.
- Supabase is appearing in the next year's top 10.

## Implications

- PostgreSQL remains the leader among open source databases and continues to grow in importance.
- SQLite is growing in embedded and lightweight scenarios, while cloud databases such as Supabase are becoming more important.
- Oracle is losing relevance and is increasingly being replaced by modern cloud and open source solutions.



# DASHBOARD

---

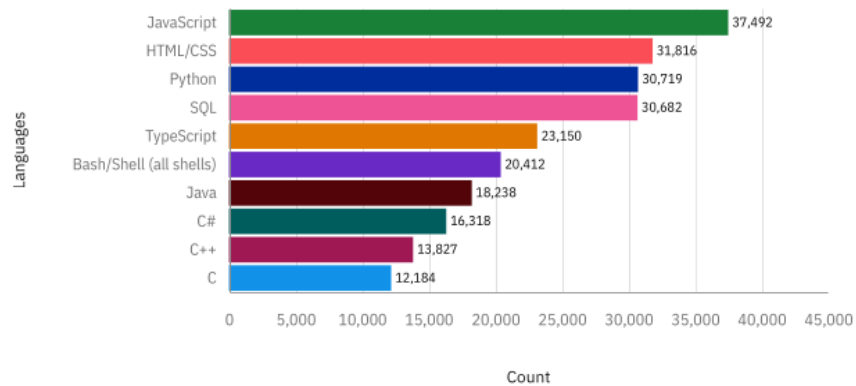


[https://github.com/Sabrinata/ibm\\_data/blob/main/IBM%20CapstoneCognos.pdf](https://github.com/Sabrinata/ibm_data/blob/main/IBM%20CapstoneCognos.pdf)

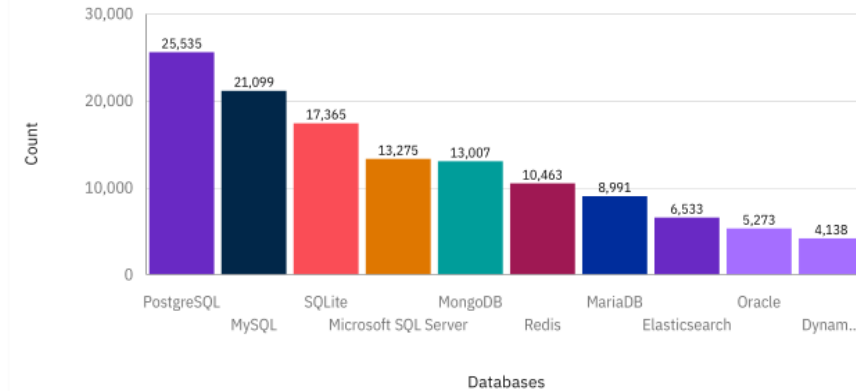
# DASHBOARD TAB 1

## Current Technology Usage

Top 10 Language Respondents Have Worked With



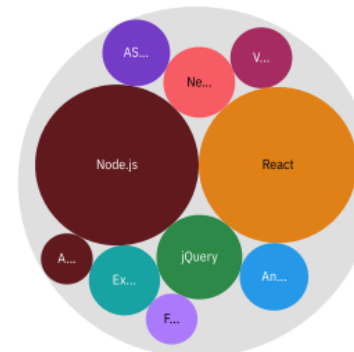
Top 10 Databases Respondents Have Worked With



Top 10 Platforms Respondants Have Worked With



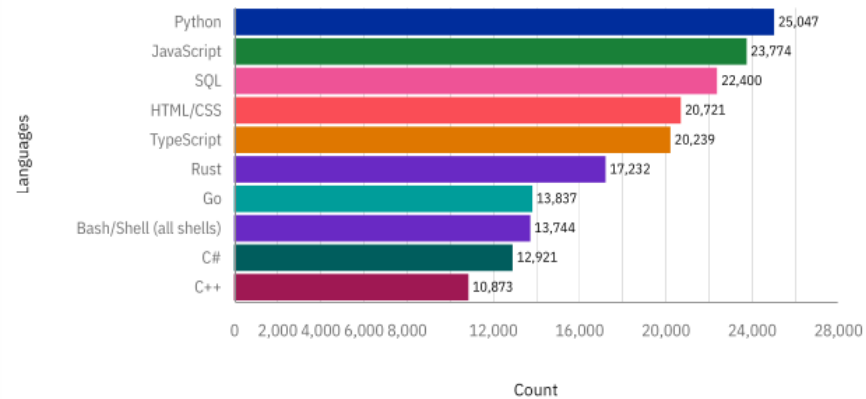
Top 10 WebFrames Respondants Have Worked With



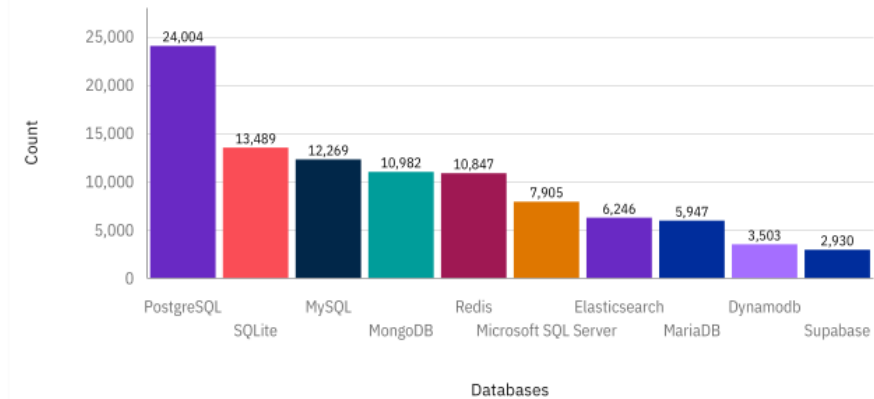
# DASHBOARD TAB 2

## Future Technology Trend

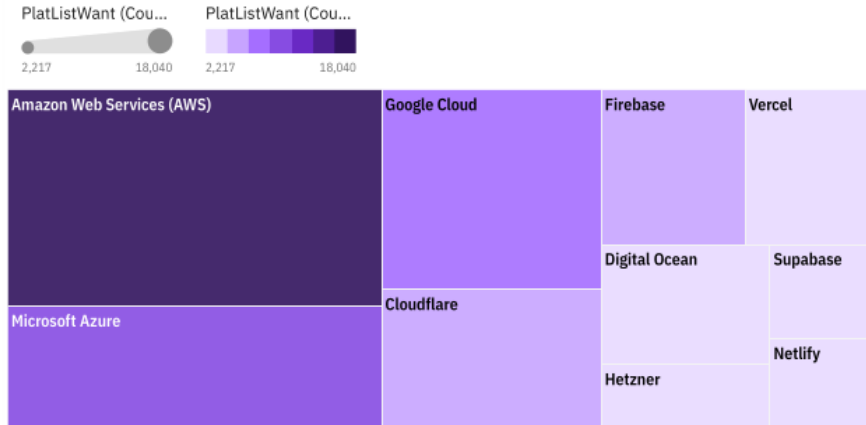
Top 10 Languages Respondents want to work with



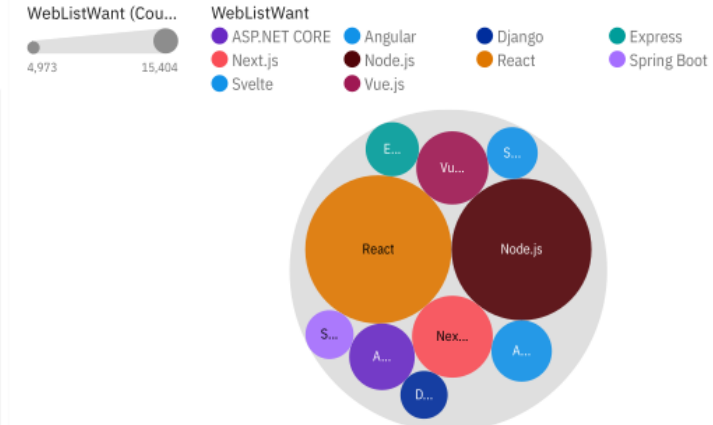
Top 10 Databases Respondents want to work with



Top 10 Platforms Respondents want to work with



Top 10 Webframes Respondents want to work with



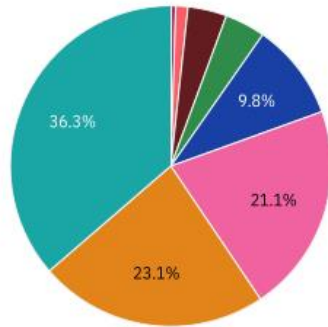
# DASHBOARD TAB 3

## Demographics

### Respondent distribution by Age

Age

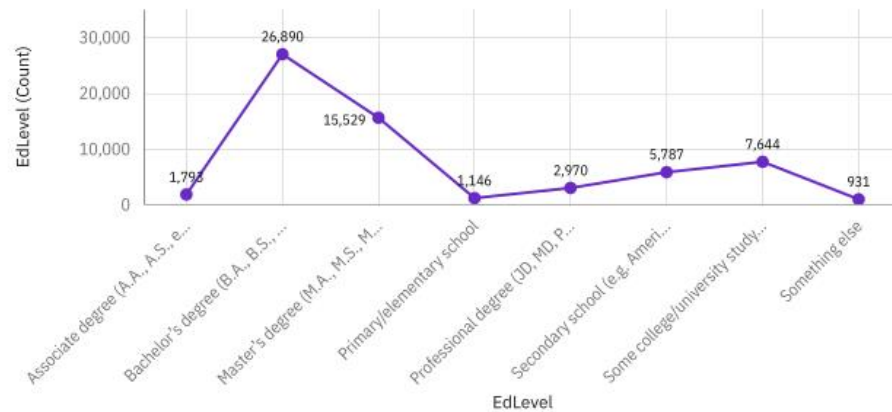
- Prefer not to say
- 18-24 years old
- 65 years or older
- Under 18 years old
- 55-64 years old
- 45-54 years old
- 35-44 years old
- 25-34 years old



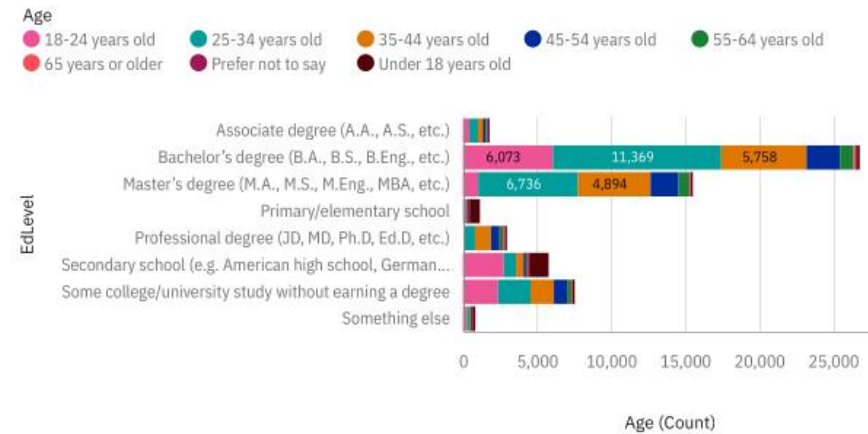
### Respondent Count by Country



### Respondent distribution by Formal Education Level



### Respondent Count by Age, classified by Education Level



# DISCUSSION

---



- **Languages**

JavaScript, HTML/CSS, and Python dominate usage. Python is expected to replace JavaScript as the most desired language.

- **Databases**

PostgreSQL leads and will strengthen its lead, followed by MySQL and SQLite. SQLite will rise to second place.

- **Platforms & Web Frames**

Cloud platforms (Amazon Web Service, Azure, Google Cloud) and modern frameworks (React, Node.js, Next.js) remain in high demand.

- **Demographics**

Majority of respondents are aged 25–34, with most holding a bachelor's degree.

# OVERALL FINDINGS & IMPLICATIONS

---

## Findings

- Python is likely to overtake JavaScript in popularity and Rust and Go are becoming increasingly important
- PostgreSQL leads and is gaining share; SQLite rises to second place; Supabase enters the top 10; Oracle drops out.
- High demand for AWS, Azure, Google Cloud, and modern frameworks like React, Node.js, and Next.js.
- Majority are younger professionals (25–34) with higher education.

## Implications

- Trends show that modern, open, and cloud-based technologies are becoming increasingly important.
- New languages and services point to a shift toward powerful, cloud-based development.
- The target group prefers simpler, more flexible, and more user-friendly solutions.



# CONCLUSION

---

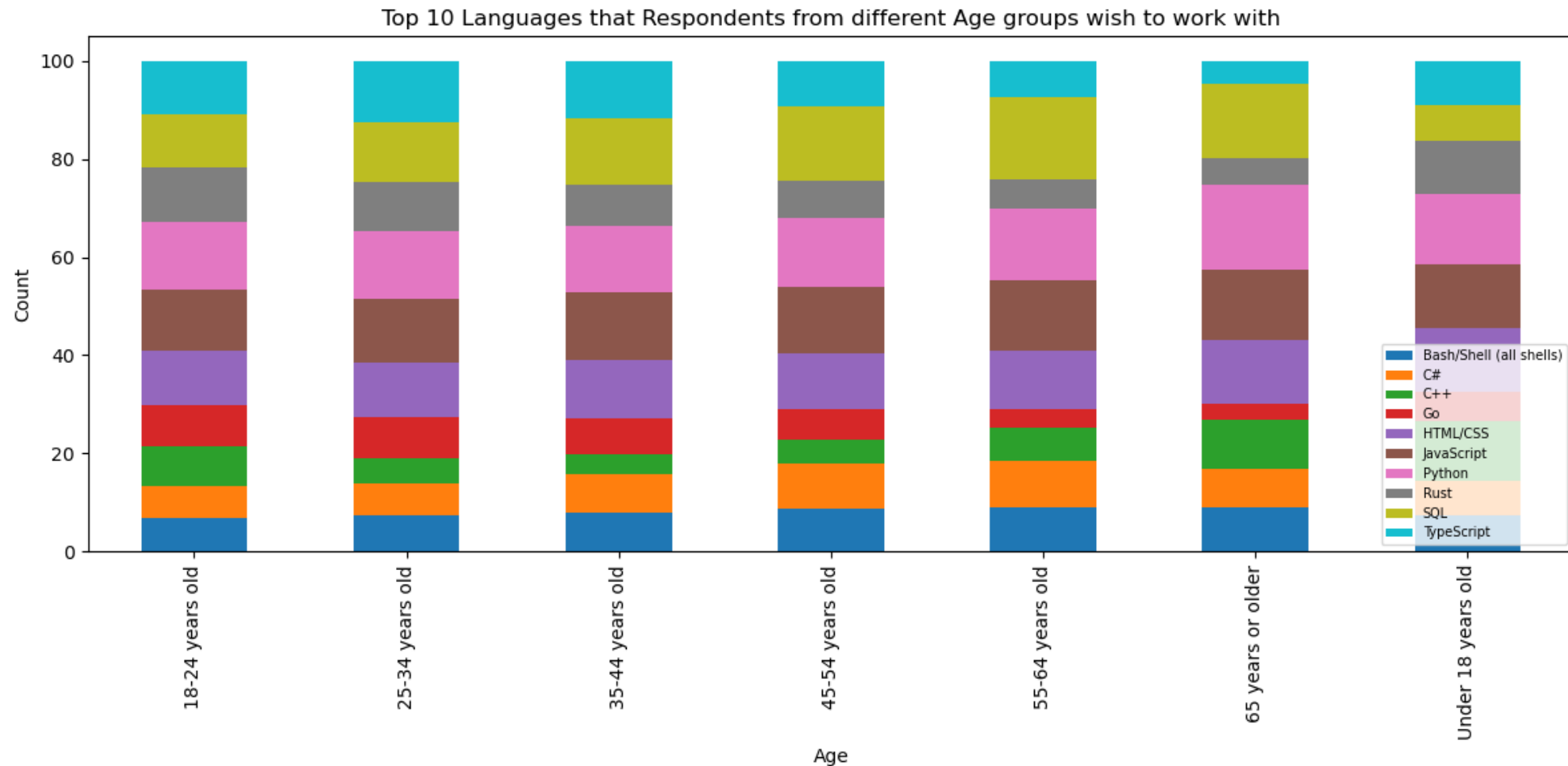


- Demand is growing for modern, open, cloud-based technologies like Python, PostgreSQL, and modern frameworks.
- Rust, Go and Supabase signal a trend toward high-performance, cloud-native development.
- Oracle, C, and older technologies are losing ground to simpler, more flexible solutions.
- A young, well-educated group of respondents is driving the rapid introduction of new technologies.

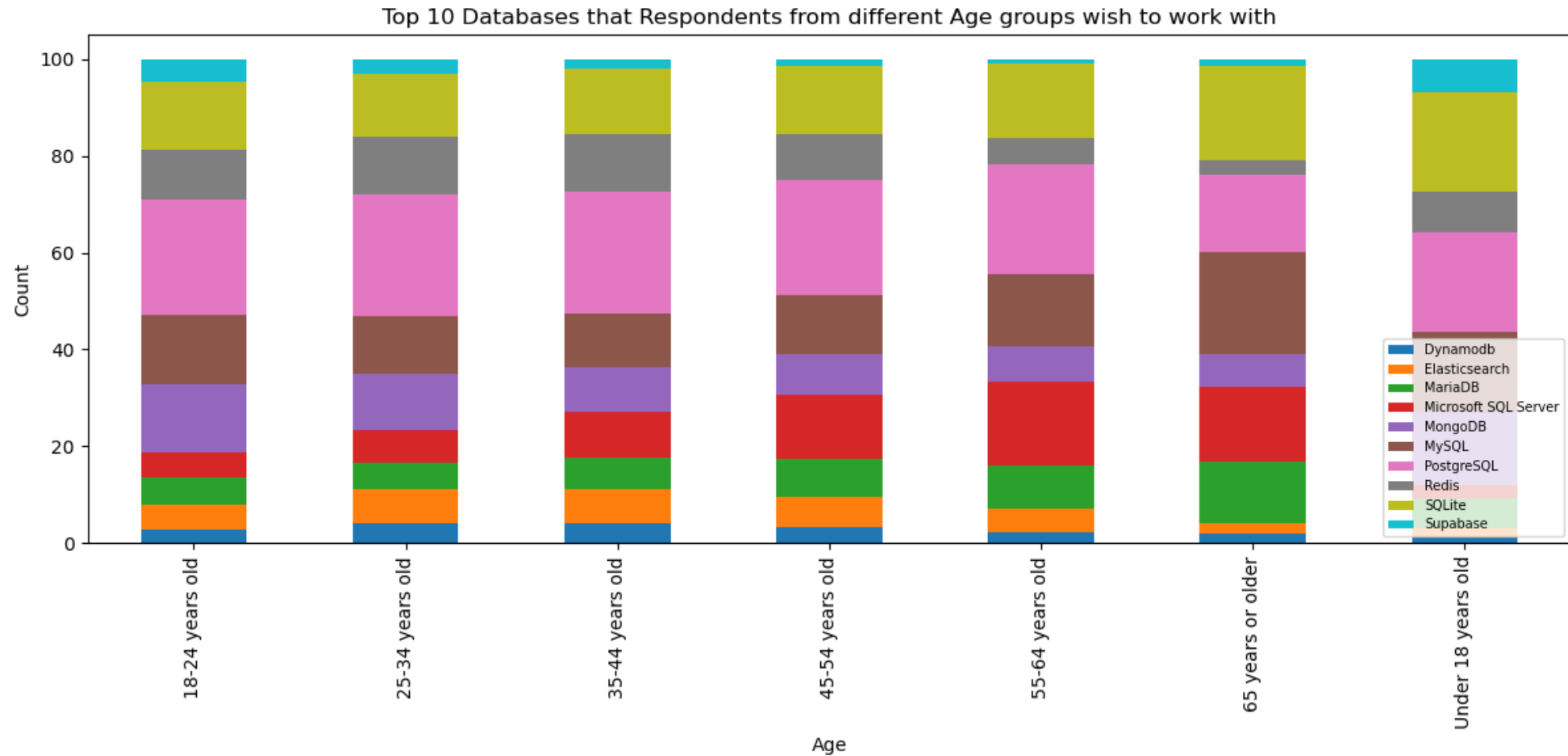




# APPENDIX

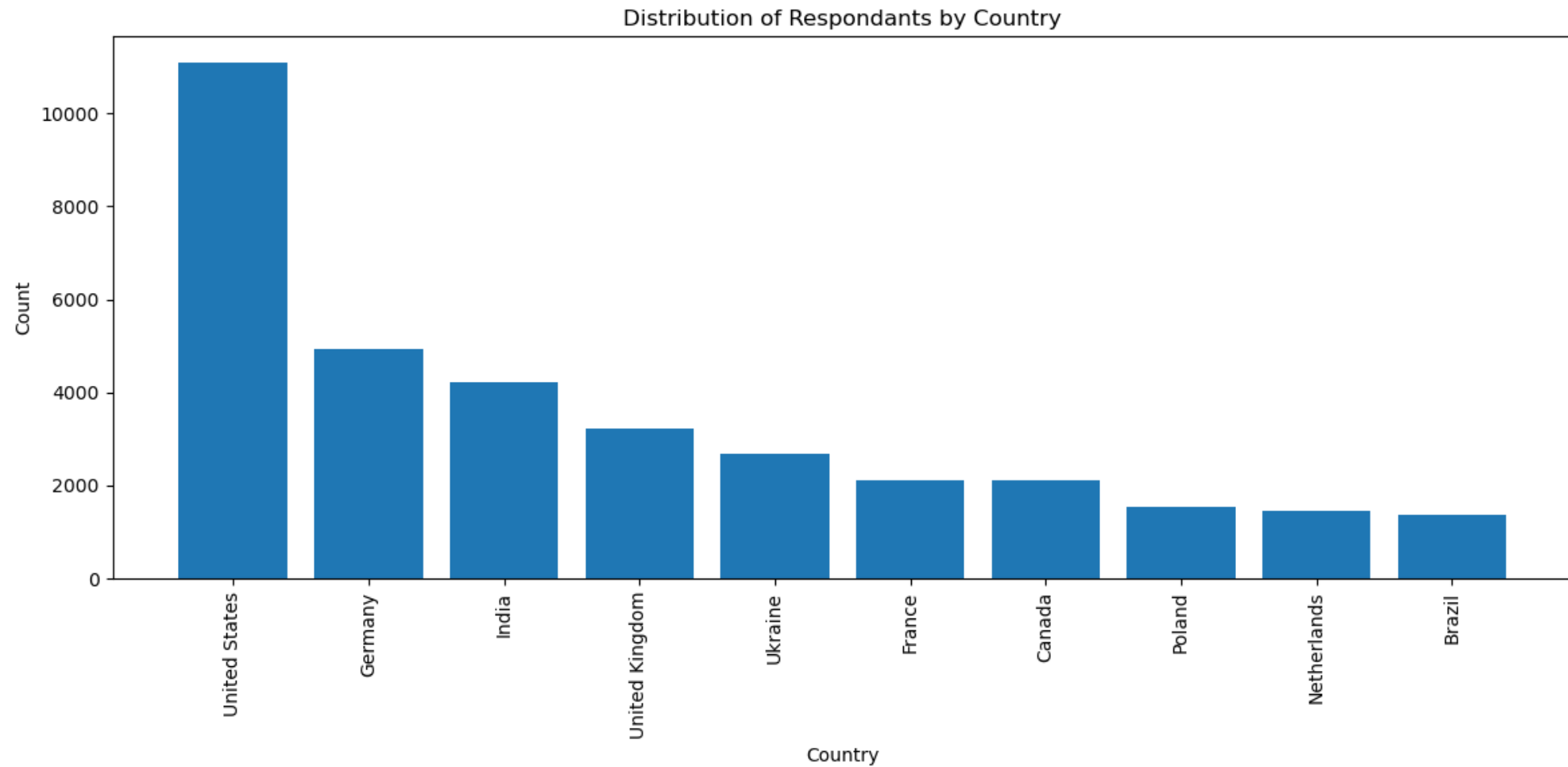


# APPENDIX



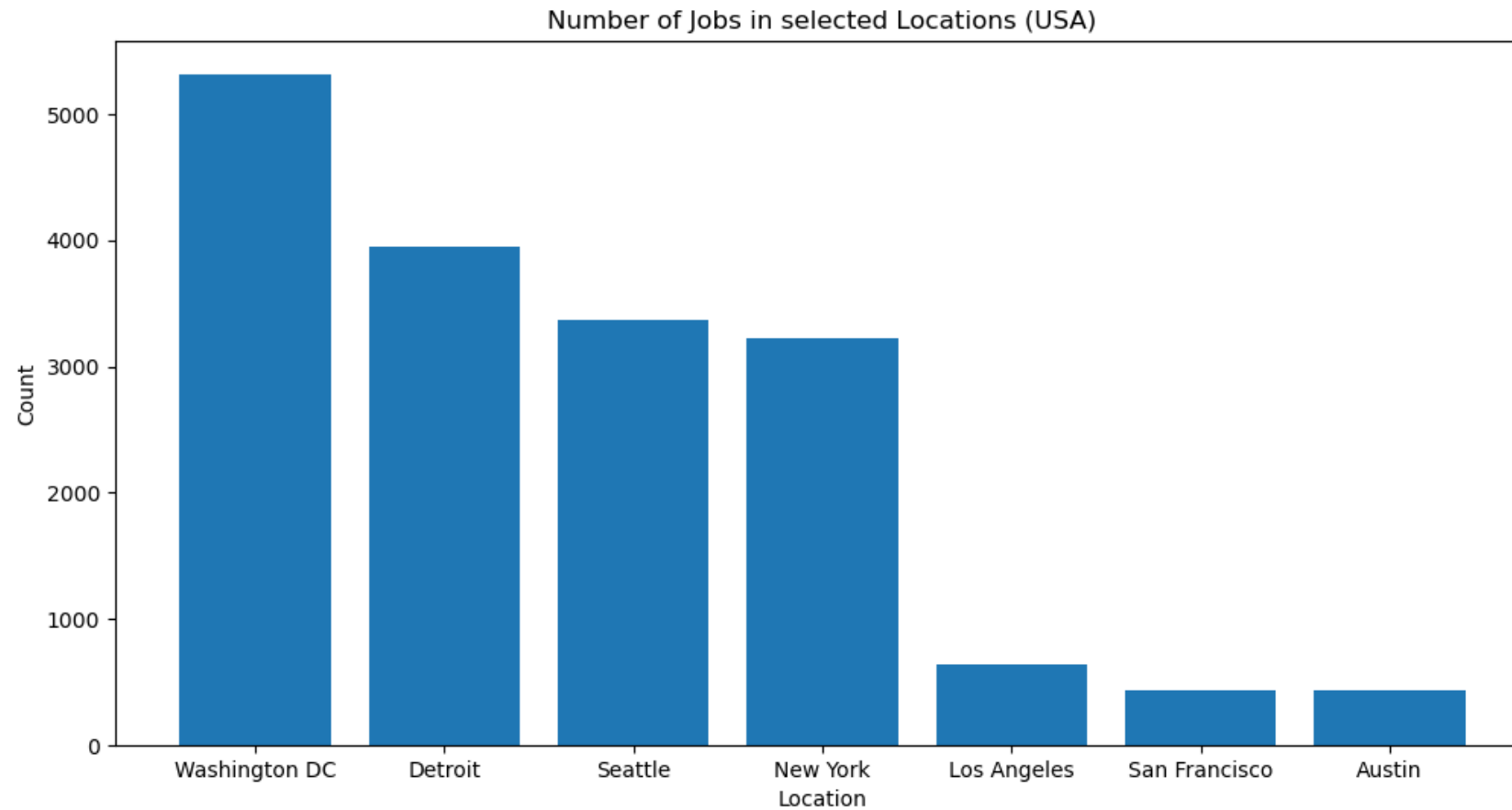
# APPENDIX

---



# JOB POSTINGS

---



# POPULAR LANGUAGES

---

